Question 4

Which speaker system could produce the loudest sound for CD Track 1?

A. speaker 1

B. speaker 2

C. speaker 3

D. speaker 4

2 marks

Question 5

Which speaker system could reproduce CD Track 2 with the best fidelity?

A. speaker 1

B. speaker 2

C. speaker 3

D. speaker 4

2 marks

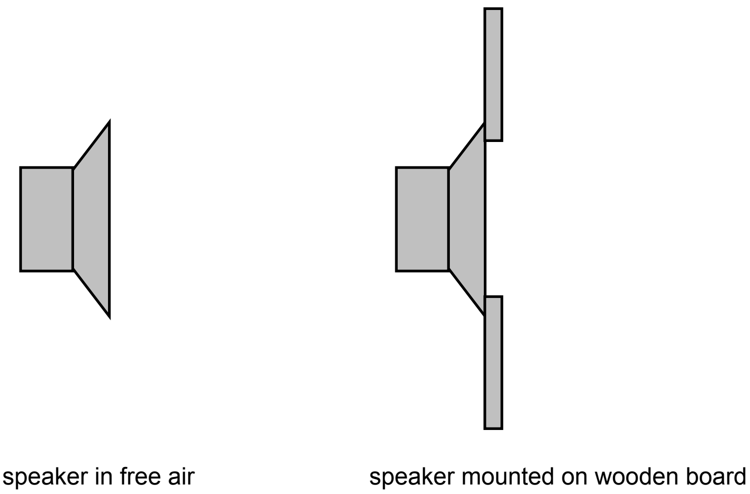

Figure 3

Question 6

A test of each speaker is performed. In each case, the volume of sound produced by the speaker increased when the speaker was mounted on a wooden board as shown in Figure 3 above. Why does this increase occur?

2 marks

Question 7

The combination of multiple speakers in an enclosure can be used to optimise the overall frequency response of a system. Identify the best **two** of the speakers (1–4), in Figure 1 page 33, for inclusion in an infinite baffle enclosure. Explain the reasoning behind your choice.

Choice 1

Choice 2

4 marks

The didgeridoo is a resonant cavity. It can be modelled as a tube of length L, open at one end and closed at the other, as shown in Figure 4. The fundamental frequency of this particular tube is 80 Hz.

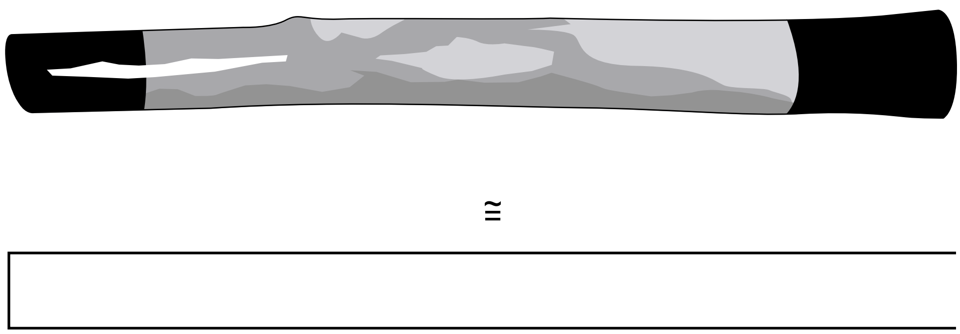

Figure 4

Question 8

Estimate the length of this tube, given the speed of sound in air is 340 m s^{-1} and a fundamental resonance of 80 Hz.

m

2 marks

Question 9

At which **one or more** of the frequencies could the tube resonate?

A. 20

B. 210

C. 320

D. 560

E. 800

2 marks

Jane and Jill have recently purchased a surround sound system for their plasma TV. They have installed the small satellite speakers (ss) as shown in Figure 5. The instructions state that the central speaker reproduces most of the dialogue (300–3000 Hz) and the sub-woofer reproduces the low-frequency sounds below 100 Hz.

Initially the central speaker is placed behind the TV screen. Jane and Jill are in the middle of the room as shown.

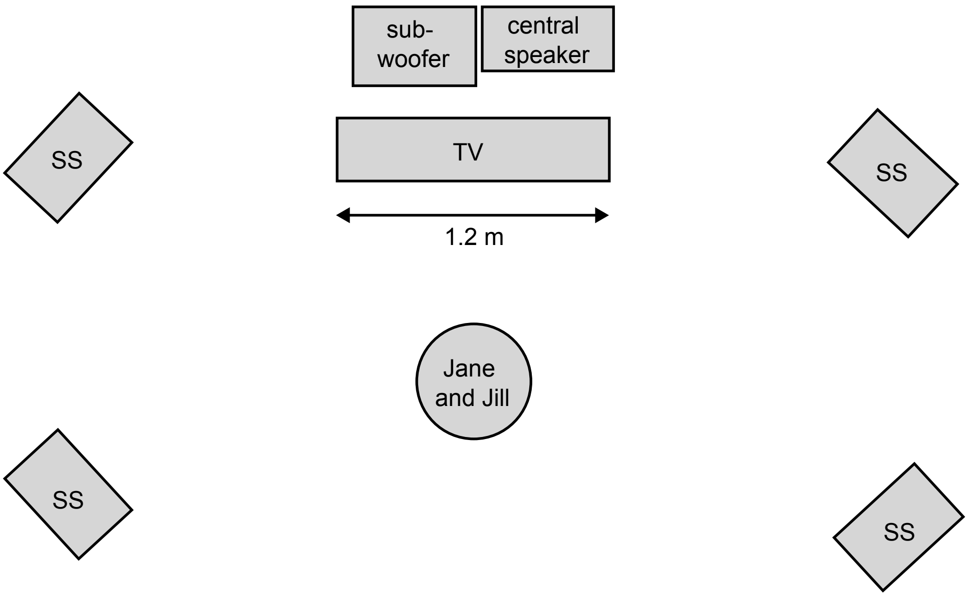

Figure 5

With the central speaker and sub-woofer behind the screen, the dialogue sounds muffled, lacking definition in the high frequencies, yet the low-frequency sounds can be heard quite clearly. When Jane repositions the central speaker underneath the screen, all frequencies can be heard clearly.

Question 10

a. Calculate the wavelength of a 100 Hz sound in air.

m

2 marks

b. Explain why the high-frequency sounds cannot be heard clearly when the central speaker is behind the screen, although the low-frequency sounds from the sub-woofer can be heard.

2 marks

END OF QUESTION AND ANSWER BOOK